

QSignAI: Quantum-Randomness-Seeded Identity Signatures at the Intersection of AI for Science and Science for AI

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Abstract—The 2024–2025 Nobel and Turing awards recognised artificial intelligence and quantum science in the same breath — machine learning as a physical science, artificial intelligence solving 50-year scientific problems, superconducting quantum circuits as the hardware foundation of quantum computing, and quantum information principles as computing’s highest achievement. Yet no deployed artificial intelligence system has brought these two streams together for the general public: identity systems still rely on pseudo-random tokens, and quantum circuits remain invisible to the billions of people who use bot-enabled social messaging platforms daily. This paper presents QSignAI, a production-deployed open-source platform demonstrating a bidirectional relationship between artificial intelligence and quantum science in a real-time event participation system. We address three research questions: first, can quantum-randomness generation via real quantum circuits be embedded in an artificial-intelligence-driven social platform with acceptable latency and cost; second, can an artificial intelligence bot make quantum phenomena perceptually legible to general audiences with no prior technical knowledge; and third, does a system combining both directions work in practice. A conversational artificial intelligence bot routes each participant’s first message through a two-circuit quantum pipeline on a cloud quantum simulator, producing a unique quantum-randomness-seeded identity signature per participant. The first two questions are answered through system design and qualitative deployment evidence; measurable comparisons are identified as priority future work.

Index Terms—quantum randomness, Bell state, identity signatures, ToyLWE, AWS Braket, conversational AI bot, AI for science, science for AI, real-time social display, post-quantum cryptography

I. INTRODUCTION

In 2024–2025, the Nobel and Turing committees recognised artificial intelligence (AI) and quantum science simultaneously. The 2024 Nobel Prize in Physics went to Hopfield and Hinton [1] for foundational discoveries enabling machine learning with artificial neural networks — the first time the Nobel Committee in Physics recognised AI research. The 2024 Nobel Prize in Chemistry recognised Hassabis, Jumper, and Baker [2] for AlphaFold2, the canonical demonstration of

AI for Science. The 2025 Nobel Prize in Physics recognised Clarke, Devoret, and Martinis [3] for macroscopic quantum tunnelling and energy quantisation in electrical circuits — the direct physical foundation of today’s superconducting quantum computers. The 2025 ACM Turing Award — computing’s highest honour — recognised Bennett and Brassard [4] for establishing the foundations of quantum information science, including the principle that quantum measurement produces randomness no classical adversary can reproduce.

Despite this convergence, three gaps persist. *Gap 1 (Science for AI)*: AI identity systems still use pseudo-random number generators (PRNGs) — deterministic algorithms that are theoretically reversible. Bennett and Brassard’s Turing Award-winning insight has not been applied to AI participation systems. *Gap 2 (AI for Science)*: quantum circuits remain invisible to non-specialists. AlphaFold2 showed AI can make science accessible at scale [2]; can an AI bot do the same for quantum science? *Gap 3 (AI for Better Life)*: transformation only reaches people when it is accessible. A bot on a platform billions already use — zero install, zero technical knowledge — is a concrete answer to how quantum science improves everyday life.

This paper addresses three Research Questions (RQs):

- **RQ1 (Science for AI)**: Can quantum-randomness generation via real quantum circuits be embedded in an AI-driven social participation system with acceptable latency, cost, and reliability?
- **RQ2 (AI for Science)**: Can an AI bot make quantum phenomena — superposition, entanglement, Bell state measurement — perceptually legible to general audiences with no prior technical knowledge?
- **RQ3 (Deployment)**: Does a system combining both directions work in practice, and what does successful deployment demonstrate about the bidirectional relationship?

The contributions of this paper are as follows. We present a complete, production-deployed system demonstrating the bidirectional AI–quantum relationship, including quantum circuit design comprising a 4-qubit random number generator (RNG)

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Open-source implementation: <https://github.com/QuantBlockchain/QSignAI>.

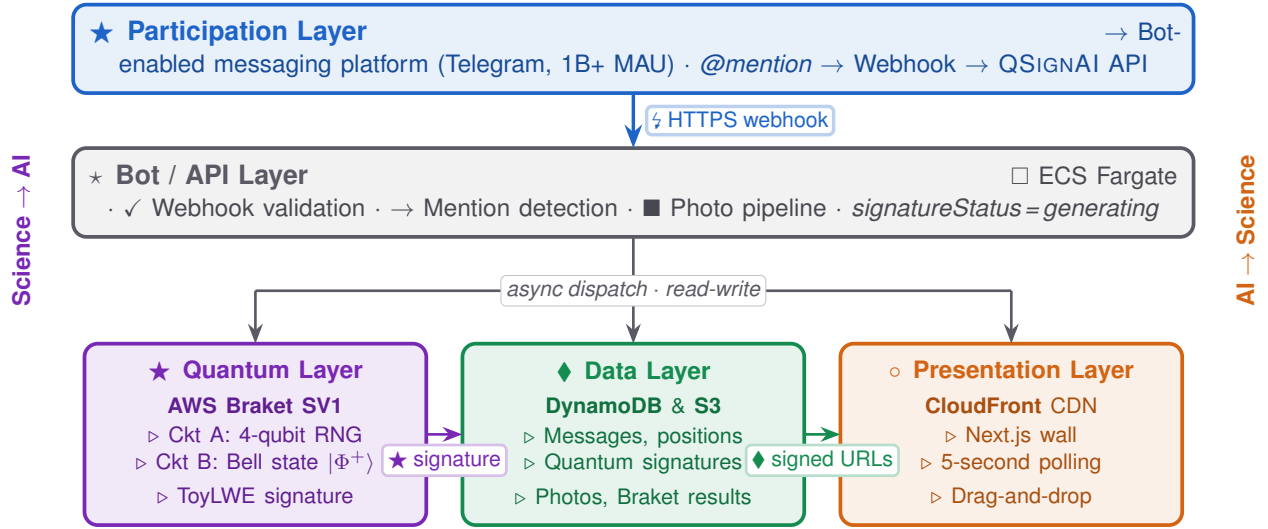


Fig. 1. System architecture of QSignAI showing the bidirectional relationship between AI and quantum science. **Layers:** ■ **Participation** (bot-enabled messaging, Telegram 1B+ MAU); ■ **Bot/API** (ECS Fargate, webhook, photo pipeline); ■ **Quantum** (AWS Braket SV1 → ToyLWE signatures); ■ **Data** (DynamoDB + S3); ■ **Presentation** (CloudFront, Next.js wall). **Directions:** Sci. → AI (quantum randomness strengthens identity); AI → Sci. (bot makes quantum legible to audiences).

plus 2-qubit Bell state on Amazon Web Services (AWS) Braket SV1 (state-vector simulator), seeding ToyLWE (Toy Learning-With-Errors) identity signatures. We describe the Telegram Bot API integration architecture with multi-group isolation and graceful degradation, and a three-surface User Experience (UX) making quantum-derived identity legible to general audiences. The open-source implementation is available at [GitHub URL].

II. BACKGROUND AND RELATED WORK

A. Quantum Randomness

Classical PRNGs are deterministic: given the same seed, they always produce the same sequence. A quantum measurement is governed by the Born rule — outcomes are determined by probability amplitudes, not hidden variables, as experimentally confirmed by Aspect et al. [5] closing the Bell inequality loophole established by Bell [6]. Cloud-accessible quantum randomness is now available via services such as AWS Braket. NIST SP 800-90B [7] defines randomness requirements for cryptographic applications. The key distinction in this paper: quantum randomness is the *input* (what the circuits produce); identity signatures are the *output* (what ToyLWE derives).

B. Post-Quantum-Inspired Signatures

The NIST post-quantum cryptography (PQC) standardisation process [8] produced CRYSTALS-Dilithium (FIPS 204 [9]), based on the Learning With Errors (LWE) problem introduced by Regev [10]. LWE mixes a secret vector with a noise vector, producing a distribution computationally indistinguishable from uniform. ToyLWE (Toy Learning-With-Errors) as used in this work is a demonstrative, pedagogical instantiation — structural analogy to LWE, not a production primitive. The path to production is replacement with CRYSTALS-Dilithium using the same quantum-randomness-seeded entropy pipeline.

C. Bot-Enabled Social Messaging Platforms

Bot-enabled group messaging platforms expose a programmable webhook application programming interface (API) enabling AI agents to mediate between human participants and backend systems. The current deployment uses Telegram, which surpassed 1 billion monthly active users (MAU) in March 2025 [11]. Prior bot applications span education, healthcare, and civic engagement [12], but none combine quantum circuits with real-time social display. Existing interactive event platforms (Slido, Mentimeter) use PRNG tokens with no quantum provenance.

D. AI for Science

The AlphaFold2 result [2] established the template for AI for Science: an AI system making a scientific problem accessible at a scale classical methods cannot match. QSignAI applies the same template to quantum science communication: an AI bot making quantum phenomena accessible to a live audience of non-specialists. Quantum machine learning [13] and Noisy Intermediate-Scale Quantum (NISQ) era computing [14] provide the broader context for near-term quantum applications.

III. SYSTEM ARCHITECTURE

The overall architecture is shown in Fig. 1. The system comprises three integrated layers.

The **Participation Layer** is a bot-enabled group messaging platform. The current deployment uses Telegram’s Bot API¹ with webhook delivery validated by X-Telegram-Bot-API-Secret-Token. Mention detection parses message.entities for type: "mention" with offset/length extraction; caption_entities handles photo messages. Photos are downloaded via the platform file API (20 MB limit) and

¹Telegram Bot API: <https://core.telegram.org/bots/api>

TABLE I
QSIGNAI REST API ENDPOINTS. ICONS DENOTE ROLE: ★ INGEST,
◆ DISPLAY, × DELETE, ○ LAYOUT, ● ADMIN, ▷ HEALTH.

Method	Endpoint	Auth	Role
POST	/api/webhook/[gld]	Token	★ Ingest
GET	/api/messages/[gld]	Public	◆ Fetch wall
DELETE	/api/messages/[gld]	Bearer	× Soft-delete
PATCH	/api/messages/[gld]	Bearer	○ Position
GET	/api/groups	Public	★ List groups
POST	/api/admin	Passwd	● Admin login
GET	/api/health	Public	▷ Health

Row color encodes HTTP verb: ■ POST (write), ■ GET (read), ■ DELETE, ■ PATCH (update). All via CloudFront TLS. *gld* = groupId.

uploaded to a private, encrypted Amazon Simple Storage Service (S3) bucket.² The **Quantum Identity Layer** runs on AWS Braket SV1³ and is described in Section IV. The **Presentation Layer** is a Next.js⁴ application served via Amazon CloudFront, polling for new messages every five seconds.

Infrastructure. The full stack is defined as infrastructure-as-code using AWS Cloud Development Kit (CDK).⁵ Elastic Container Service (ECS) Fargate runs 1–4 container instances, auto-scaling at 70% CPU utilisation. Amazon DynamoDB (DynamoDB) stores messages with composite key GROUP#{groupid} / MSG#{timestamp}#{messageld}, encrypted at rest with point-in-time recovery enabled. All credentials are stored in AWS Secrets Manager; no secrets appear in code or container images. Scaling parameters are summarised in Table II.

Security. Multi-layer security includes CloudFront secret-header validation blocking direct Application Load Balancer (ALB) access, Telegram token validation on every webhook request, bearer token authentication for admin endpoints, HTTP Strict Transport Security (HSTS) and Content Security Policy (CSP) headers, least-privilege Identity and Access Management (IAM) roles, and HTML entity encoding for all user input (4096-character maximum). The system API endpoints are listed in Table I.

IV. QUANTUM RANDOMNESS PIPELINE

A. Why Quantum Randomness

A PRNG is a deterministic function: given the same seed, it always produces the same output. An adversary who recovers the seed can reproduce every identity token the system has ever issued. A quantum measurement has no such algorithm underneath it. The Born rule states that measurement outcomes are governed by probability amplitudes — not hidden variables, not prior system state. This was experimentally confirmed by Aspect et al. [5]. Quantum-randomness-seeded identity tokens cannot be predicted or reproduced by any classical adversary,

²AWS S3: <https://docs.aws.amazon.com/s3/>

³AWS Braket: <https://docs.aws.amazon.com/braket/>

⁴Next.js: <https://nextjs.org/>

⁵AWS CDK: <https://docs.aws.amazon.com/cdk/>

TABLE II
ECS FARGATE AUTO-SCALING PARAMETERS. VALUE COLORS ENCODE
SCALE: ■ MINIMAL/STABLE, ■ MODERATE.

Parameter	Value	Rationale
Min instances	1	Always-on for webhook
Max instances	4	Handle event spikes
CPU target	70%	Scale before saturation
Scale-out cooldown	30 s	Respond to traffic bursts
Scale-in cooldown	60 s	Avoid flapping
Memory	1024 MB	Next.js + image processing
CPU	0.5 vCPU	Sufficient for I/O workload

Managed by AWS Application Auto Scaling. Min. one instance keeps webhook always available.

regardless of computational power, because the randomness was not computed — it was measured. This is the direct applied instantiation of the principle recognised by the 2025 ACM Turing Award [4].

B. Circuit A: 4-Qubit Quantum RNG

The two quantum circuits are shown in Fig. 2. Circuit A is specified in Open Quantum Assembly Language (OpenQASM) 3.0 and submitted to AWS Braket SV1. Hadamard gates place all four qubits into superposition simultaneously. A Controlled-NOT (CNOT) chain entangles them: the measurement outcome of any qubit is correlated with all others. User-seeded rotations $R_y(\theta_i)$ bias the probability distribution without determining the outcome, where

$$\theta_i = \frac{\pi \cdot (\text{charCode}(\text{username}[i]) \bmod 128)}{128}. \quad (1)$$

One hundred measurement shots are taken; the most-frequently observed bitstring is converted to an integer:

$$q_{\text{num}} = \text{int}(\text{topBits}, 2) \bmod 1001, \quad (2)$$

yielding $q_{\text{num}} \in [0, 1000]$.

C. Circuit B: 2-Qubit Bell State Measurement

Circuit B creates the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ [5], [6]. Two hundred measurement shots yield the probability distribution $\beta = [P(|00\rangle), P(|01\rangle), P(|10\rangle), P(|11\rangle)]$. In an ideal simulator, $P(|00\rangle) \approx P(|11\rangle) \approx 0.5$; deviations introduced by the user-seeded rotations provide a second independent entropy source and serve as a structural witness of quantum entanglement embedded in every participant’s badge. The circuit parameters for both circuits are summarised in Table III.

D. ToyLWE Signature Derivation

The quantum number seeds a ToyLWE-inspired signature using SHAKE-256 (Secure Hash Algorithm 3 extendable-output function) [15]:

$$\mathcal{S} = \text{SHAKE-256}(\mathcal{U} \parallel q_{\text{num}} \parallel r), \quad (3)$$

$$\mathcal{H} = \text{SHA-256}(\mathcal{S}[0:32])[0:12], \quad (4)$$

$$\mathcal{G} = \text{SHA-256}(\mathcal{H}_{\text{msg}} : \mathcal{H}_{\text{ent}} : \mathcal{H}), \quad (5)$$

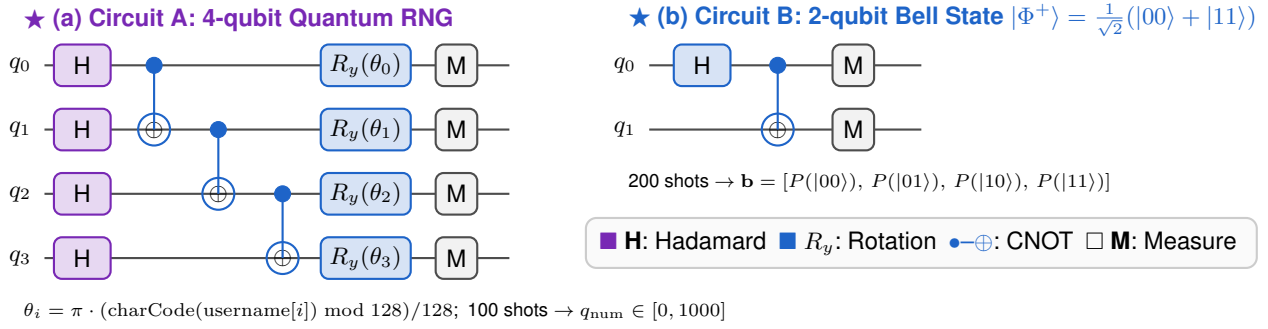


Fig. 2. Quantum circuits used in QSignAI. (a) **Circuit A** (4-qubit RNG): ■ H gates create superposition; ■ R_y rotations personalise the distribution; 100 shots $\rightarrow q_{\text{num}} \in [0, 1000]$. (b) **Circuit B** (Bell state $|\Phi^+\rangle$): ■ H + CNOT create maximal entanglement; 200 shots $\rightarrow \beta = [P(00), \dots, P(11)]$, encoding quantum entanglement in each participant’s card colour.

TABLE III
QUANTUM CIRCUIT PARAMETERS. ■ CIRCUIT A (RNG); ■ CIRCUIT B (BELL STATE).

Parameter	Circuit A (RNG)	Circuit B (Bell)
Qubits	4	2
Gate sequence	H, CNOT chain, R_y	H, CNOT
Shots	100	200
Output	$q_{\text{num}} \in [0, 1000]$	$\beta = [P(00), \dots, P(11)]$
Purpose	Quantum random seed	Entropy + visual colour
Exec. time	2–8 s (SV1)	2–8 s (SV1)

Both circuits are specified in OpenQASM 3.0 and executed on AWS Braket SV1. Results are written to an S3 bucket with prefix *amazon-braket**.

where r is 32 fresh random bytes preventing replay, $\mathcal{H}_{\text{msg}}$ is the SHA-256 hash of the message content, and $\mathcal{H}_{\text{ent}}$ is the SHA-256 hash of q_{num} . The public key hash $\mathcal{H}_{\text{pk}}$ (12 uppercase hex characters) and 24-character base64 signature form the badge displayed on each card: $Q_{452} | 7B284BB3D413$. The structural analogy to LWE [10] is that q_{num} plays the role of the secret and SHAKE-256 mixing plays the role of noise addition. This is a demonstrative instantiation; the path to production is replacement with CRYSTALS-Dilithium [9].

E. Visual Identity Encoding

The quantum number and Bell state probabilities are mapped to a Hue-Saturation-Lightness (HSL) colour value:

$$h = (q_{\text{num}} \times 137.5) \bmod 360, \quad (6)$$

$$s = 70 + \beta_0 \times 30, \quad (7)$$

$$l = 45 + \beta_3 \times 20, \quad (8)$$

where $\beta_0 = P(|00\rangle)$ and $\beta_3 = P(|11\rangle)$. The golden-angle increment (137.5°) maximises perceptual separation between consecutive hue values, a technique from phyllotaxis. The degree of quantum entanglement is encoded in the vividness and brightness of the card colour: the colour of a card *is* the quantum data, making quantum science perceptually legible without any technical knowledge.

F. Graceful Degradation

A 30-second timeout or Braket task failure triggers a local SHAKE-256-seeded fallback. Affected rows are flagged *algorithm*: “ToyLWE-local-fallback” and *device*: “local-fallback”. The wall never blocks; UX continuity is preserved. The admin dashboard exposes the device field per row, making quantum execution provenance auditable.

V. USER EXPERIENCE WALKTHROUGH

The three-surface interface is shown in Fig. 3. The interface follows a three-surface guided flow.

Surface 1 (Messaging Platform): participants use their existing messaging client — zero installation required. The @mention acts simultaneously as a routing signal and a lightweight opt-in consent gate. Only mention-bearing messages are persisted.

Surface 2 (Public Photo Wall): the wall polls for new messages every five seconds. Each card displays the sender’s name, message text, optional photo (click-to-zoom lightbox), and the quantum badge. Card colour is derived from Bell state probabilities via (6)–(8), making quantum entanglement visible to a live audience. Participants drag cards to preferred positions; positions are persisted as viewport percentages in DynamoDB, surviving window resizes and refreshes. A leaderboard ranks top contributors by message count.

Surface 3 (Admin Dashboard): event organisers access a password-protected dashboard exposing per-row quantum provenance (device, algorithm, signatureStatus, bellState, quantumNumber). Soft-delete sets `hidden = true` rather than removing rows, preserving the audit trail and quantum signatures.

The UX arc moves from *invisible* (quantum circuits running asynchronously in the background) to *visible* (Bell state probabilities encoded as card colour on the public wall) to *auditable* (quantum execution provenance inspectable by organisers). This arc is the AI-for-Science contribution: quantum phenomena made legible to a live audience through a familiar social interface.

The complete end-to-end interactive flow across all three surfaces is shown in Fig. 4, tracing a single participant’s

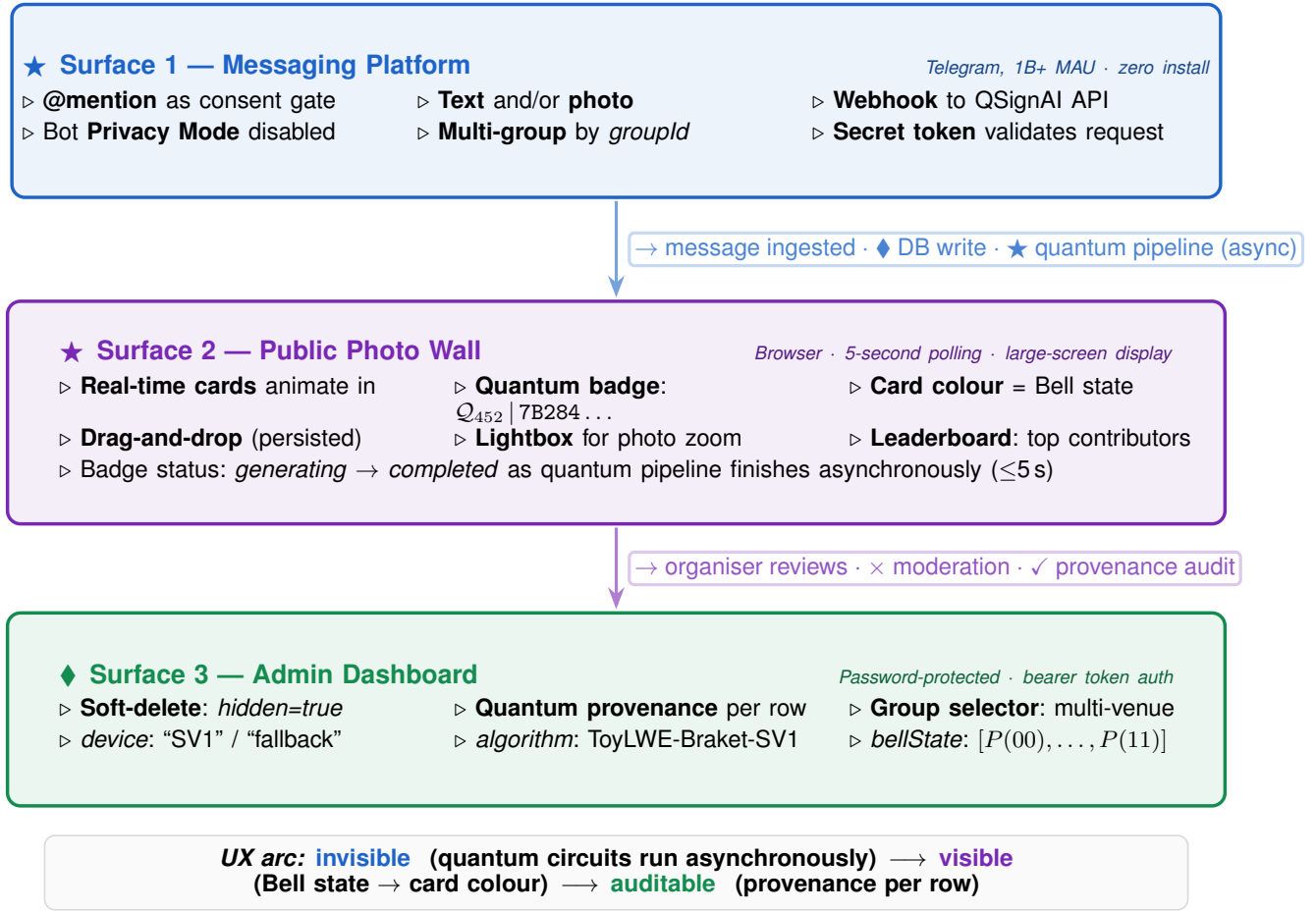


Fig. 3. Three-surface UX architecture of QSignAI. ■ **Surface 1:** bot-enabled messaging (Telegram, 1B+ MAU), @mention as consent gate; ■ **Surface 2:** public photo wall with quantum badge and Bell-state card colour; ■ **Surface 3:** admin dashboard with per-row quantum provenance audit. UX arc: invisible → visible → auditable.

message from submission through the quantum pipeline to the public wall and admin dashboard.

VI. DEPLOYMENT AND DISCUSSION

A. Answering RQ3: What Successful Deployment Demonstrates

RQ3 asks whether a system combining quantum randomness and AI bot mediation works in practice. The deployment answers this in three ways. *First, feasibility:* the system has been deployed on standard AWS infrastructure, processed real event participants, and operated without blocking on quantum task latency — demonstrating that quantum circuits are compatible with live event timing constraints when the pipeline is designed asynchronously. *Second, bidirectional value:* without the quantum circuits, the bot’s identity tokens are indistinguishable from any PRNG-based system. Without the bot, the quantum circuits produce outputs that no general audience ever sees. The combination creates a system with a property neither component achieves alone. *Third, graceful degradation:* the fallback mechanism demonstrates that quantum enhancement can be added to an AI system without introducing a single point of failure.

B. Deferred Measurable Comparisons

The deployment establishes feasibility qualitatively. Measurable comparisons deferred to future work include: a rigorous NIST SP 800-90B [7] comparison of quantum-randomness-seeded versus PRNG-seeded token distributions; systematic latency benchmarks across SV1, IonQ Aria, Rigetti Ankaa, and IBM Eagle; a controlled user study measuring whether exposure to quantum-derived visual badges builds quantum literacy; and a formal security comparison of ToyLWE versus CRYSTALS-Dilithium [9] under the LWE hardness assumption [10].

C. Simulator Versus Physical QPU

AWS Braket SV1 is a classical simulation of quantum behaviour — its randomness is ultimately pseudo-random at the hardware level. True quantum randomness requires a physical quantum processing unit (QPU). The architecture is QPU-ready: the device Amazon Resource Name (ARN) in the quantum signature module is the only required change. The 2025 Nobel Prize in Physics [3] recognised the hardware foundations that make physical QPUs possible; integrating one is the primary near-term extension.

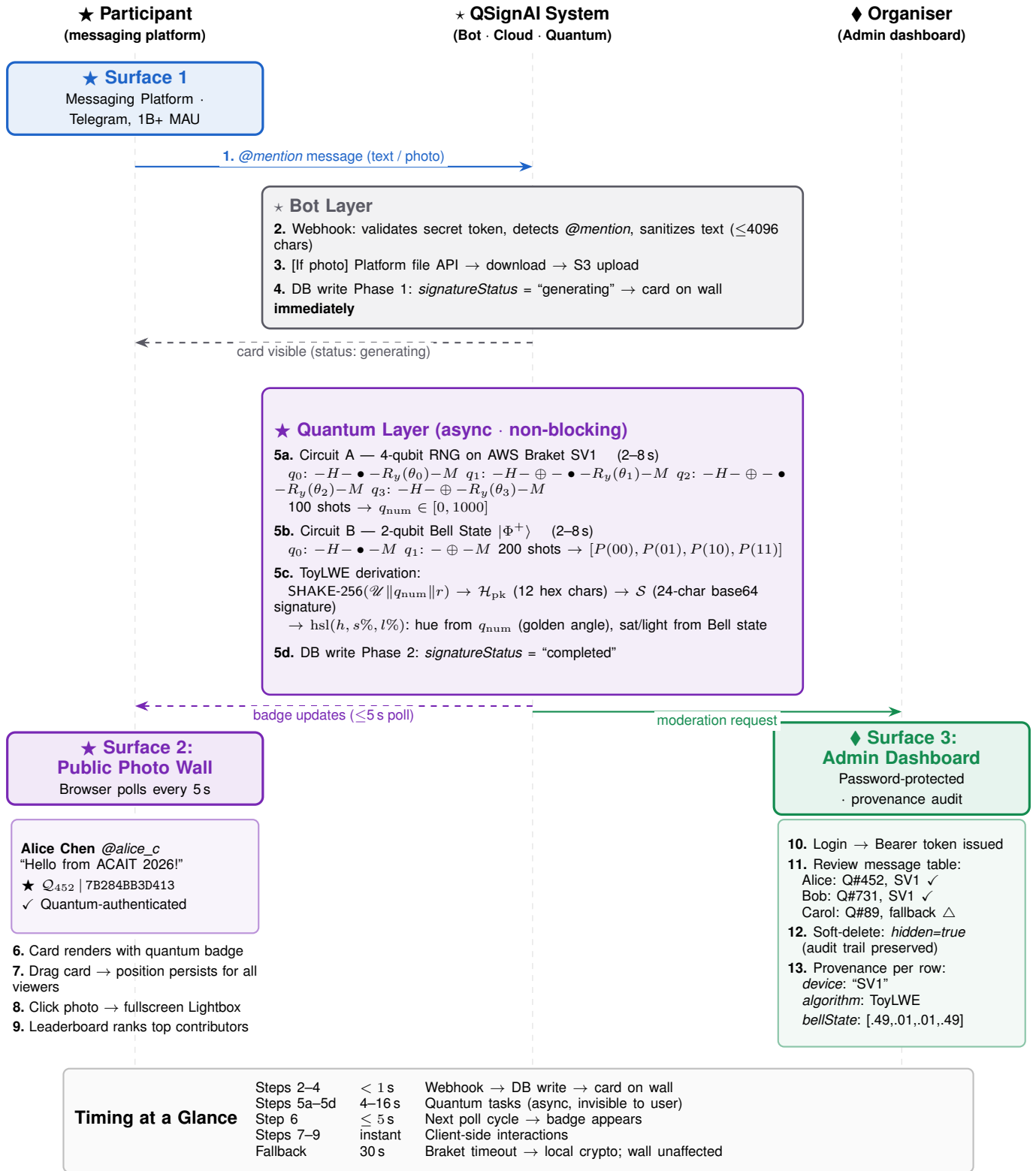


Fig. 4. Complete interactive flow of QSignAI across three surfaces. **Actors:** ★ Participant (messaging platform user); ★ QSignAI System (Bot, Cloud, Quantum); ◆ Organiser (Admin dashboard). **Surfaces:** ■ Surface 1 (messaging platform, Telegram 1B+ MAU); ■ Surface 2 (public photo wall, 5-second polling); ■ Surface 3 (admin dashboard, provenance audit). Steps 2–4 complete in < 1 s; quantum tasks (Steps 5a–5d) run asynchronously in 4–16 s; the badge updates on the next poll cycle (≤ 5 s). The fallback path (30 s Braket timeout) preserves UX continuity. See Appendix A for term definitions.

D. AI for Better Life Alignment

QSignAI addresses the conference theme on three axes. *Accessibility*: quantum-authenticated identity is delivered through a channel over one billion people already use [11], with no new hardware, no new application, and no technical knowledge required. *Transparency*: every participant's identity has verifiable quantum provenance, auditable by organisers through the admin dashboard. *Scientific literacy*: quantum concepts — superposition, entanglement, Bell states — are made perceptually legible to live audiences via visual badges, aligning with Sustainable Development Goal (SDG) 4 (Quality Education) [16]. The system also aligns with SDG 9 (Industry, Innovation and Infrastructure), SDG 16 (Peace, Justice and Strong Institutions), and SDG 17 (Partnerships for the Goals).

VII. FUTURE RESEARCH ECOSYSTEM

QSignAI is the deployed foundation for a broader research ecosystem, illustrated in Fig. 5.

Near-term (1–2 years): Physical QPU integration will replace SV1 with IonQ Aria, Rigetti Ankaa, or IBM Eagle for genuine hardware quantum randomness, with NIST SP 800-90B certification [7]. CRYSTALS-Dilithium replacement [9] will use the same entropy pipeline with production-grade PQC signatures and a formal security proof under the LWE hardness assumption [10]. TON blockchain on-chain identity anchoring will issue Soulbound Tokens (SBTs) as non-transferable Non-Fungible Token (NFT) event credentials via TON Connect. A Model Context Protocol (MCP) server [17] will expose quantum-authenticated participant data as an MCP resource, natively queryable by Large Language Model (LLM) agents.

Mid-term (2–4 years): Agentic wallet contracts will enable AI-initiated on-chain credential issuance. Cocoon Trusted Execution Environment (TEE)-based confidential compute will support privacy-preserving analytics over participant data. Reinforcement Learning (RL) [18] will power adaptive moderation agents.

Long-term (4+ years): Zero-Knowledge (ZK) proofs over post-quantum signatures, longitudinal quantum literacy studies, and cross-platform quantum identity standards represent the long-horizon research agenda.

VIII. CONCLUSION

QSignAI demonstrates that the relationship between AI and quantum science is bidirectional: quantum randomness, rooted in the principles recognised by the 2025 ACM Turing Award [4] and enabled by the hardware recognised by the 2025 Nobel Prize in Physics [3], strengthens AI-driven identity (Science for AI, RQ1); and the AI bot layer, following the template established by AlphaFold2 [2], makes quantum phenomena accessible to general audiences at event scale (AI for Science, RQ2). Neither direction is theoretical — both are demonstrated in a single production-deployed system (RQ3). The central finding: a conversational AI bot and two quantum circuits on a cloud simulator are sufficient to make quantum science tangible to over one billion potential users today [11]. QSignAI is not the destination; it is the foundation.

ACKNOWLEDGMENT

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APPENDIX

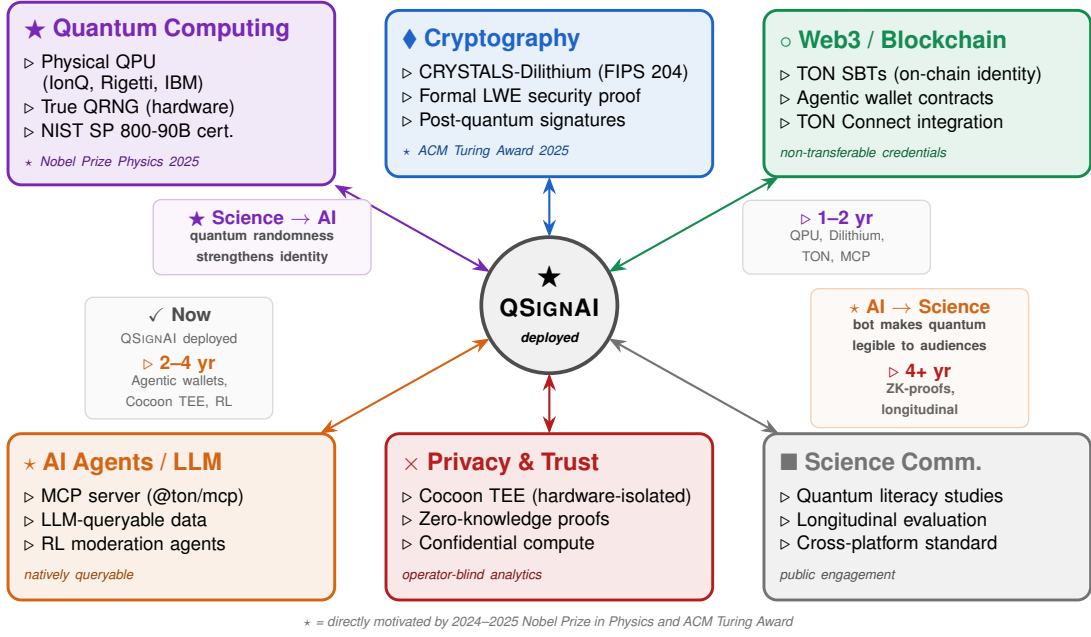


Fig. 5. Future research ecosystem with QSIGNAI as the deployed foundation, arranged as a 3×2 grid of research domains. **■ Quantum Computing** (physical QPU, ★Nobel Prize 2025); **■ Cryptography** (CRYSTALS-Dilithium, ★Turing Award 2025); **■ Web3** (TON SBTs, agentic wallets); **■ AI Agents** (MCP server, RL moderation); **■ Privacy** (Cocoon TEE, ZK-proofs); **■ Sci. Comm.** (quantum literacy studies). The timeline strip shows the four-horizon research roadmap. ★ = motivated by 2024–2025 Nobel/Turing awards.

TABLE IV

GLOSSARY OF KEY TERMS, ABBREVIATIONS, AND NOTATION (SEE ALSO
INLINE DEFINITIONS AT FIRST USE IN THE TEXT).

Term	Definition	Term	Definition
QRNG	Quantum RNG: physically irreducible randomness	PRNG	Pseudo-RNG: deterministic; theoretically reversible
SV1	AWS Braket state-vector simulator (classical)	QPU	Quantum Processing Unit: physical hardware
CNOT	Controlled-NOT: 2-qubit entangling gate	Bell state	$ \Phi^+\rangle = (00\rangle + 11\rangle)/\sqrt{2}$
OpenQASM	Open Quantum Assembly Language	ToyLWE	Toy LWE: demonstrative lattice signature
LWE	Learning With Errors: lattice hard problem	PQC	Post-Quantum Cryptography
SHAKE-256	SHA-3 extendable-output function	HSL	Hue-Saturation-Lightness colour model
q_{num}	Quantum number $\in [0, 1000]$ from Circuit A	β	Bell state vector from Circuit B
$\mathcal{S}$	Seed: SHAKE-256($\mathcal{U} q_{\text{num}} r$)	$\mathcal{H}$	Public key hash: first 12 hex chars
$\mathcal{G}$	Signature: 24-char base64 string	$\mathcal{U}$	Username seed for R_y rotations
AWS	Amazon Web Services: cloud platform	ECS	Elastic Container Service
DynamoDB	Amazon managed NoSQL database	S3	Amazon Simple Storage Service
UX	User Experience: three-surface design	MAU	Monthly Active Users
MCP	Model Context Protocol [17]	LLM	Large Language Model
SBT	Soulbound Token: non-transferable NFT	TEE	Trusted Execution Environment
ZK	Zero-Knowledge proof	RL	Reinforcement Learning [18]
SDG	Sustainable Development Goal [16]	RNG	Random Number Generator

Terms grouped by domain. All circuit notation follows OpenQASM 3.0. Inline definitions at first use in the main text.